

Online Energy and Interference Management for Dynamic Target Tracking With Cellular-Connected UAV

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Abstract—Cellular-connected Unmanned Aerial Vehicles (UAVs) have significant potential for target tracking in future cellular networks due to their broad coverage and operational flexibility. In this paper, we consider a multi-cell cellular network with a cellular-connected UAV for target tracking, which encounters challenges such as unpredictable flight energy consumption from the stochastic movements of the tracking target and severe uplink interference from ground devices (GDs). To tackle these challenges, we propose a multi-stage stochastic optimization framework focused on energy-efficient target tracking with interference coordination. Our objective is to optimize the long-term average uplink throughput of both aerial users and GDs by jointly optimizing the UAV's trajectory, power allocation, and cell association across multiple orthogonal communication resource blocks (RBs). The formulated stochastic non-convex problem is first transformed into a deterministic problem for each time slot by using the Lyapunov optimization framework. An online optimization strategy is proposed, utilizing the optimal structure, alternative optimization, and successive convex approximation (SCA) techniques. Simulation results show that the proposed approach significantly enhances network throughput and UAV energy queue stability compared to existing baseline schemes.

Index Terms—Cellular connected UAV, dynamic target tracking, energy management, Lyapunov optimization.

I. INTRODUCTION

CELLULAR-CONNECTED Unmanned Aerial Vehicles (UAVs) are becoming key elements in the advancement of future cellular networks as novel aerial users with a variety of applications, such as remote sensing, cargo transportation, and aerial target tracking [1], [2], [3], [4]. Unlike traditional UAVs with limited visual line-of-sight (VLoS) communications, cellular-connected UAVs improve remote communication and control by enhancing reliability, security, and throughput through the use of existing high-capacity cellular backhaul links. Recent studies have leveraged these UAVs for aerial target tracking in regional monitoring to ensure public safety. The work in [5] employed UAVs for ocean target tracking using a lightweight energy-saving scheme, which optimized real-time tracking with an Elman neural network, thus reducing energy consumption while ensuring high tracking success and data stability. The work in [6] proposed a cooperative multi-agent reinforcement learning (MARL) framework for regional target tracking to enhance tracking efficiency and accuracy through optimizing power consumption and extending UAV lifetimes.

The challenge in UAV tracking arises from the stochastic nature of target movements, which makes energy consumption unpredictable. This unpredictability arises from several factors, including sudden changes in the UAV's trajectory and varying speeds required to maintain target tracking. Traditional deterministic methods are insufficient for managing such uncertainties in practical scenarios. The work in [7] proposed an online learning strategy for optimizing task completion delays in UAV-enabled mobile edge computing (MEC) systems, where dynamic user movements and task arrivals were addressed using reduced complexity methods. The work in [8] proposed online computation offloading strategies in MEC systems, where task execution delay and energy consumption were optimized by leveraging nearby access points (APs) and server selection algorithms under communication dynamics. In [9], UAVs in a MEC network optimized energy consumption by dynamically allocating resources and trajectories through a velocity-triggered penalty term without considering the air-to-ground interference.

Integrating UAVs into cellular networks presents fundamental challenges, particularly due to potential interference. The high altitude of UAVs creates a line-of-sight (LoS) communication channel with base stations (BSs), which offers both benefits and

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challenges. While LoS channels ensure steadier communication and enable UAVs to link with multiple BSs for network diversity enhancement, they also complicate interference management by increasing potential uplink interference across a broader set of BSs. Prior works [10], [11], [12] have explored various strategies for interference mitigation in UAV networks, including full-dimensional antenna arrays and multicell beamforming. However, these works have primarily focused on interference coordination technologies, overlooking the unpredictability of energy consumption in random target tracking flights. Integrating such unpredictability with cellular-connected UAVs presents several challenges. First, the UAVs must effectively track randomly moving targets, which requires dynamic adaptability. Second, it is essential to manage energy consumption efficiently to prolong UAV operational times. Lastly, resource allocation must be controlled to minimize air-to-ground interference while ensuring balanced throughput for both aerial user and ground devices (GDs).

To tackle these challenges, in this paper, we develop a design framework for cellular-connected UAV target tracking. This framework addresses the unpredictability of flight energy consumption by considering the stochastic nature of target movements while incorporating interference coordination for throughput improvement with both aerial users and GDs. The main contributions of this paper are summarized as follows:

- We develop a multi-stage stochastic optimization framework for interference-aware cellular-connected UAV target tracking to enhance uplink throughput for both aerial and ground users while ensuring energy efficiency. This is achieved by simultaneously optimizing UAV tracking trajectory, BS association, and power allocation across multiple orthogonal resource blocks (RBs).
- We first transform the formulated problem into a deterministic model for each time slot by employing Lyapunov optimization techniques. By treating energy constraint as virtual energy-aware queue dynamics and balancing average network throughput and energy consumption with a penalty factor, we develop a comprehensive framework that effectively manages the tradeoff between communication performance and energy efficiency.
- We propose an efficient online strategy that decouples the deterministic optimization problem into subproblems for UAV trajectory design, transmit power control, and BS association optimization. These subproblems are iteratively solved by optimal structure analysis and successive convex approximation (SCA) techniques.
- Simulation results show that compared to the baseline schemes, the proposed solution effectively reduces interference between aerial users and GDs to improve overall network throughput. Additionally, the results show that the proposed solution ensures energy efficiency and enhances system stability.

The rest of this paper is organized as follows. Section II presents related work. Section III describes the system model and formulates the optimization problem. In Section IV, we present the Lyapunov optimization based online control algorithm. In Section V, we address the single time slot problem.

TABLE I
LIST OF NOTATIONS

Notation	Definition
$\mathcal{J}$	The set of BSs
$\mathcal{N}$	The set of time slots
$\mathcal{M}$	The set of all RBs
$\mathbf{s}_j^{\text{BS}}$	The horizontal coordinate of BS j
$\mathbf{s}_n^{\text{tar}}$	The horizontal coordinate of the target at time slot n
$\mathbf{s}_{j,n}^{\text{UAV}}$	The horizontal coordinate of the UAV at time slot n
κ	The target randomness level
v_n^{tar}	The speed of the target at time slot n
θ_n^{tar}	The direction of the target at time slot n
$\bar{v}$	The target mean speed and direction
$\bar{\theta}$	The target mean direction
d_n^{tar}	The distance between the UAV and the target at time slot n
$d_{j,n}^{\text{UAV}}$	The distance between the UAV and BS j at time slot n
D	The tracking range
$\mathcal{N}_q^j$	The set of first q -tier neighboring cells for BS j
$H_{j,m,n}$	The GD-BS channel coefficient for BS j with RB m at time slot n
$G_{j,m,n}^{\text{UAV}}$	The UAV-BS channel coefficient for BS j with RB m at time slot n
$\hat{p}_{j,m,n}$	The transmit power of the GD for BS j with RB m at time slot n
β_0	The channel power gain
α	The path loss exponent
$\sigma_{j,m,n}^2$	The total background noise power for BS j with RB m at time slot n
$R_g(m,n)$	The G2G throughput for RB m at time slot n
$R_u(m,n)$	The A2G throughput for RB m at time slot n
$Q(n)$	The virtual energy queue
$\mathbf{v}_n^{\text{UAV}}$	The flight velocity of the UAV at time slot n
$P^{\text{f}}(n)$	The flight power during flight at time slot n
$E^{\text{f}}(n)$	The flight energy consumption at time slot n
E^{max}	The maximum allowable average energy consumption
V	The penalty factor

Simulation results are presented in Section VI. Finally, conclusions are drawn in Section VII. The main notations used throughout the paper are summarized in Table I.

II. RELATED WORK

In this section, we review the existing literature on cellular-connected UAVs, focusing on aspects such as target tracking, interference management, and energy consumption.

A. Target Tracking With Cellular-Connected UAVs

Cellular-connected UAVs have been extensively studied for their ability to leverage existing cellular infrastructure for communication, control, and target tracking across various applications. In [13], a two-phase command and control transmission scheme was proposed. This scheme optimizes message relays under latency and energy constraints using a decentralized constrained graph attention multi-agent Deep-Q-network approach. A three dimensional (3D) path planning framework for cellular-connected UAVs was proposed in [14], which optimizes flying distance while maintaining communication quality with ground BSs using signal-to-interference-plus-noise ratio (SINR) maps and grid quantization. Furthermore, the authors in [15] examined coordinated multi-point non-orthogonal multiple access (CoMP-NOMA) integration in cellular-connected UAV networks to enhance reliability and average ergodic rates for both aerial and terrestrial users.

Target tracking using cellular-connected UAVs has gained significant attention due to its potential in surveillance, search and rescue, and disaster management. In [16], a cooperative mission strategy was proposed using UAVs for a search-and-track mission of underwater targets, which optimizes path planning considering maneuverability and communication constraints.

The work in [17] proposed a deep reinforcement learning (DRL) approach for multi-UAV tracking of first responders in complex 3D environments for enhanced tracking accuracy. Furthermore, the authors in [18] proposed a DRL-based deep deterministic policy gradient (DDPG) model for visual UAV tracking, optimizing energy efficiency and tracking accuracy. In [19], [20], methods were proposed to enhance UAV target tracking, collision avoidance, and trajectory prediction using UAV swarm-based cooperative architecture and distributed model predictive control (DMPC)-based approaches. The work in [21] presented a framework for optimizing the energy-efficient 3D trajectory of a solar-powered UAV monitor for autonomous tracking of suspicious UAVs. However, these studies overlook the practical challenges posed by interference and energy issues in cellular-connected UAVs.

B. Interference Management for Cellular-Connected UAVs

Interference management is essential for cellular-connected UAVs, especially in densely populated urban areas. In [22], a novel UAV enabled reconfigurable intelligent surface (RIS)-aided interference management scheme for space-air-ground integrated networks was proposed, which leverages various types of channel state information (CSI) and employs interference alignment, beamforming, and space-time precoding to improve system capacity. The study in [23] developed an alternating-maximization approach to optimize 3D UAV trajectories and transmission powers, aiming to maximize data throughput while mitigating interference from existing communication networks. In [24], an optimization algorithm was proposed for joint IRS-user association, UAV trajectory optimization, successive interference cancellation (SIC) decoding order scheduling, and power allocation to maximize energy efficiency in multi-IRS UAV communications. Additionally, the work in [25] proposed an interference-aware path planning scheme using deep reinforcement learning to optimize energy efficiency and reduce interference in cellular-connected UAV networks. This scheme achieves adaptive UAV altitude adjustment for enhanced performance. Despite these advancements, the specific impact of interference on UAV-enabled target tracking with energy consideration remains inadequately explored.

C. Energy Issues for Cellular-Connected UAVs

Efficient energy management is a critical challenge for cellular-connected UAVs. In [26], an algorithm combining DRL and linear programming was proposed to maximize energy efficiency and ensure offloading fairness in UAV-assisted MEC by optimizing UAV trajectory, flight time, and offloading decisions of terminal devices. The research in [27] presented a 3D energy-efficient trajectory optimization framework for rotary-wing UAVs in downlink communication, which adapts to stochastic wind effects using an offline-based online adaptive approach. In [28], an energy-efficient data collection method using RIS and UAVs was proposed, optimizing UAV trajectory and sensor schedules to minimize energy consumption. Furthermore, the work in [29] explored a control-based 3D trajectory optimization method for UAV-aided wireless communication,

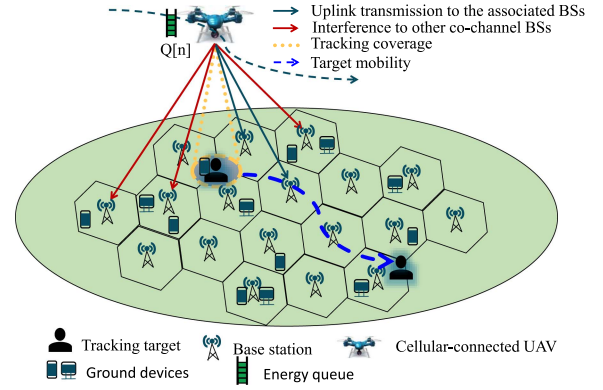


Fig. 1. Target tracking with cellular-connected UAV.

ensuring smooth trajectories and validating an improved energy consumption model through simulations. However, these studies did not consider energy consumption in the context of UAV-enabled target tracking, particularly considering the target's randomness and the air-to-ground interference environment.

Different from the above works, we investigate a comprehensive framework for scalable target tracking with random movements using cellular-connected UAVs. We address both interference management and energy efficiency by jointly optimizing UAV trajectory, power allocation, and BS association. Our method ensures reliable target tracking despite the target's randomness and minimizes interference to maximize both aerial and ground throughput. This approach conserves energy consumption, providing a robust solution for practical deployment.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. Network Model

As shown in Fig. 1, we investigate a hybrid air-to-ground (A2G) communication network with a cellular-connected UAV. In the area ξ , there are J BSs, denoted by $\mathcal{J} \triangleq \{1, 2, \dots, J\}$. Each BS j covers a certain number of GDs, denoted by K_j , where $K_j \geq 1$. Consequently, the total number of GDs in the area ξ is $K = \sum_{j=1}^J K_j$. These GDs, situated at fixed locations, continuously transmit data to their corresponding BSs through ground-to-ground (G2G) uplink channels. In addition, a cellular-connected UAV is deployed above them to track a moving target, such as a vehicle or a person of interest, where the data captured (e.g., image, video) are transmitted to ground BSs via A2G uplink channels [30], [31]. The UAV flies at a fixed altitude of H to avoid the high energy consumption associated with constant altitude variation [32]. The total flight time of the UAV is denoted as T , which is subdivided into N discrete time slots, each with a duration of $\delta = \frac{T}{N}$. These time slots are represented by the set $\mathcal{N} \triangleq \{1, 2, \dots, N\}$. The horizontal plane coordinates of j th BS is denoted as $\mathbf{s}_j^{\text{BS}} \in \mathbb{R}^{2 \times 1}$. The horizontal plane coordinates of the UAV and the tracked target at time slot n are denoted as $\mathbf{s}_n^{\text{UAV}}, \mathbf{s}_n^{\text{Tar}} \in \mathbb{R}^{2 \times 1}$, respectively. In particular, $\mathbf{s}_{(0)}^{\text{UAV}}$ and $\mathbf{s}_0^{\text{Tar}}$ denote the initial positions of the UAV and the tracked target, respectively. Let $v_{\max}$ represent the UAV's maximum flight

speed. Consequently, at each time slot, the UAV's flight distance must meet the maximum flight distance constraint, i.e.,

$$\| \mathbf{s}_{(n)}^{\text{UAV}} - \mathbf{s}_{(n-1)}^{\text{UAV}} \| \leq v_{\max} \delta, \quad \forall n \geq 1, \quad (1)$$

where $\| \cdot \|$ denotes the euclidean norm. The distance between the UAV and the BS j in time slot n is calculated as $d_{j,(n)}^{\text{UAV}} = \sqrt{H^2 + \|\mathbf{s}_{(n)}^{\text{UAV}} - \mathbf{s}_j^{\text{BS}}\|^2}$.

The terrestrial mobile target moves according to the Gauss-Markov mobility model [33], [34]. In particular, the direction and speed of the tracked target at time slot n are given by

$$v_n^{\text{Tar}} = \kappa v_{n-1}^{\text{Tar}} + (1 - \kappa) \bar{v} + \sqrt{1 - \kappa^2} v_x^{\text{Tar}}, \quad (2)$$

$$\theta_n^{\text{Tar}} = \kappa \theta_{n-1}^{\text{Tar}} + (1 - \kappa) \bar{\theta} + \sqrt{1 - \kappa^2} \theta_x^{\text{Tar}}. \quad (3)$$

In this model, $\kappa \in [0, 1]$ shows the degree of randomness in the tracked target's motion, while $\bar{\theta}$ and $\bar{v}$ denote the target's average direction and speed, respectively. Specifically, higher κ values lead to smoother trajectories that are more influenced by past states, whereas lower κ values introduce greater variability and unpredictability in the target's movement. The variables v_x^{Tar} and θ_x^{Tar} are Gaussian random variables with zero mean and unit variance. These Gaussian random variables introduce stochastic elements into the target's motion, capturing real-world uncertainties and ensuring that the instantaneous speed and direction fluctuate around the long-term mean behavior. As a result, the tracked target's position at time slot n can be calculated as

$$x_n^{\text{Tar}} = x_{n-1}^{\text{Tar}} + v_{n-1}^{\text{Tar}} \delta \cos \theta_{n-1}^{\text{Tar}}, \quad (4)$$

$$y_n^{\text{Tar}} = y_{n-1}^{\text{Tar}} + v_{n-1}^{\text{Tar}} \delta \sin \theta_{n-1}^{\text{Tar}}. \quad (5)$$

Although the Gauss-Markov model may not capture all real-world target movement complexities, it balances mathematical tractability and flexibility through the correlation factor κ , making it suitable for our optimization framework. Additionally, our framework is model-agnostic, allowing seamless integration with more sophisticated mobility models in future work to better represent real-world dynamics. The horizontal distance between the tracked target and the UAV during time slot n is denoted as $d_{(n)}^{\text{Tar}} = \|\mathbf{s}_n^{\text{Tar}} - \mathbf{s}_{(n)}^{\text{UAV}}\|$. In every time slot, the UAV maintains reliable tracking by keeping the moving target within its designated tracking radius D , which can be given by

$$(d_{(n)}^{\text{Tar}})^2 + H^2 \leq D^2, \quad \forall n. \quad (6)$$

B. Communication Model

For uplink communication resources, the total number of orthogonal RBs (e.g., frequency band) assigned is denoted by M . In practical scenarios, it is common to have $M < K$ due to the reuse of frequencies among the K GDs [35]. The notation $\mathcal{M} \triangleq \{1, 2, \dots, M\}$ represents the set of RBs. During uplink data transmission from the UAV to ground BSs, two types of interference are observed: one is A2G interference, caused by the LoS dominance of the A2G channel affecting GDs sharing the same RB; the other is terrestrial interference, resulting from the shared use of RBs among GDs. To mitigate terrestrial interference, an terrestrial interference aware RB allocation strategy is adopted wherein each BS assigns RBs to its GDs

with interference mitigation consideration [30]. Specifically, the set of first q -tier neighboring BSs of BS j is defined as $\mathcal{N}_q^j$, $q \geq 1$. Before assigning an RB to a new GD within the coverage of BS j , it is essential to verify that the RB is not allocated within the neighboring set $\mathcal{N}_q^j$. If the RB is not in use within this neighboring set, it can then be assigned to the GD, thereby minimizing potential terrestrial interference. We define a RB allocation BS set $\mathcal{J}_{m,n}$, where $j \in \mathcal{J}_{m,n}$ indicates that RB m is allocated to a GD in the cell of BS j during time slot n . This ensures that RB allocation minimizes terrestrial interference among GDs sharing the same RB by keeping track of which BSs have been assigned RB m during each time slot. On the other hand, to alleviate A2G interference, we calculate the complement set $\mathcal{J}_{m,n}^c = \mathcal{J} \setminus \mathcal{J}_{m,n}$, which represents the set of BSs not currently utilizing RB m at time slot n . The complement set $\mathcal{J}_{m,n}^c$ is essential for mitigating A2G interference as it allows the UAV to select a ground BS from the set $\mathcal{J}_{m,n}^c$ for transmission with RB m at time slot n , thereby reducing the likelihood of interference. It is worth noting that for any RB m and time slot n , the set $\mathcal{J}_{m,n}^c$ is always non-empty. The reason is that each BS, and its q -tier neighboring BSs, cannot simultaneously allocate the same RB to their respective covered GDs due to the terrestrial interference aware RB allocation strategy. Following the 3GPP standards, the modeling of G2G and A2G channels is conducted under the Urban Macro (UMa) scenario [36], [37].

1) *Ground-to-Ground Channel*: Let $k_{j,m,n}$ denote the index of the GD within the coverage of BS j , which uploads data to BS j using RB m during time slot n . The channel coefficient between GD $k_{j,m,n}$ and BS j at time slot n with RB m is represented by $H_{j,m,n}$. $H_{j,m,n}$ is influenced by several factors, including path loss, the antenna gain of the BS, and small-scale fading, which follows the Rayleigh fading model. The uplink data rate for GD $k_{j,m,n}$ at time slot n with RB m is given by:

$$R_g^j(m, n) = B \log_2 \left(1 + \frac{\hat{p}_{j,m,n} H_{j,m,n}}{\sigma_{j,m,n}^2 + p(m, n) G_{m,n}^j} \right), \quad (7)$$

where $G_{m,n}^j$ denotes the channel coefficient between the UAV and BS j for time slot n with RB m . $\hat{p}_{j,m,n}$ represents the transmit power of the GD $k_{j,m,n}$. Term $p(m, n) G_{m,n}^j$ exists in (7) due to the A2G interference affecting GDs sharing the same RB. B denotes the bandwidth allocated per RB, measured in Hertz (Hz), and $\sigma_{j,m,n}^2$ signifies the total background noise power and remaining interference from other terrestrial cells at cell j for the given time slot n with RB m . The transmit power of the UAV at time slot n with RB m is denoted as $p(m, n)$. Subsequently, the cumulative data rate of all GDs at time slot n with RB m is

$$R_g(m, n) = \sum_{j \in \mathcal{J}_{m,n}} R_g^j(m, n). \quad (8)$$

Then, the G2G throughput for all GDs at time slot n can be calculated as $\sum_{m \in \mathcal{M}} \delta R_g(m, n)$.

2) *Air-to-Ground Channel*: Note that $G_{m,n}^j$ is the channel coefficient between the UAV and BS j at time slot n with RB m . Similar to [38], the A2G channel exhibits flat frequency characteristics due to the existence of LoS propagation. Therefore,

for each time slot n , we have $G_{m,n}^j = G_n^j, \forall j \in \mathcal{J}, m \in \mathcal{M}$. We adopt a Rician fading channel model [39], [40], incorporating both LoS and non-Line-of-Sight (NLoS) components, to accurately represent A2G channels in practical urban environments where obstacles affect signal propagation. The channel coefficient G_n^j is defined as

$$G_n^j = \frac{\beta_0 |\mu_n|^2}{(H^2 + \|s_{(n)}^{\text{UAV}} - s_j^{\text{BS}}\|^2)^{\alpha/2}}, \quad (9)$$

where $\mu_n = \sqrt{\frac{K_c}{K_c+1}}\bar{g} + \sqrt{\frac{1}{K_c+1}}\tilde{g}$, with $\bar{g}$ representing the deterministic LoS component (with $|\bar{g}| = 1$) and $\tilde{g}$ denoting the random scattered NLoS component, modeled as a zero-mean unit-variance circularly symmetric complex Gaussian (CSCG) random variable. K_c is the Rician factor that quantifies the ratio of power in the LoS component to that in the NLoS components. β_0 is the channel power gain with $d_0 = 1$ m and $\alpha \geq 2$ is the pass loss component.

For each RB $m \in \mathcal{M}$ at time slot n , we denote $j(m, n)$ as the index of BS associated with the UAV for uplink transmission, where $j(m, n) \in \mathcal{J}_{m,n}^c$ is a design variable in this paper. In this case, the substantial A2G interference can be mitigated between the UAV and GDs that are covered by BSs in $\mathcal{J}_{m,n}$. To evaluate transmission reliability, we incorporate an outage probability tolerance ϵ as described in [41], [42]. The transmission rate from the UAV to BS $j(m, n)$ using RB m at time slot n is then calculated as

$$R_u(m, n) = B \log_2 \left(1 + \frac{F^{-1}(\epsilon)p(m, n)G_n^j}{\Gamma \sigma_{j(m,n),m,n}^2} \right), \quad (10)$$

where $\sigma_{j(m,n),m,n}^2$ signifies the total background noise power. $F(\cdot)$ is the cumulative distribution function (CDF) of the Rician fading coefficient while $F^{-1}(\cdot)$ is the inverse function, and Γ represents the SNR gap between practical modulation schemes and theoretical Gaussian signaling. Therefore, the A2G throughput of the UAV during time slot n is calculated as $\sum_{m \in \mathcal{M}} \delta R_u(m, n)$.

C. Energy Consumption Model

Energy consumption of the UAV includes both flight energy consumption and communication energy consumption. However, flight energy consumption is vastly greater than communication energy, often by several orders of magnitude, with flight energy typically measured in kilojoules and communication energy in mere joules [43]. Consequently, the overall energy consumption is primarily driven by the flight energy, which is the main focus of this paper. Let $\mathbf{v}_{(n)}^{\text{UAV}}$ represent the UAV's flight velocity during time slot n , where $\mathbf{v}_{(n)}^{\text{UAV}} = (\mathbf{s}_{(n)}^{\text{UAV}} - \mathbf{s}_{(n-1)}^{\text{UAV}})/\delta$. The flight power consumption during flight at time slot n is expressed as, i.e.,

$$P_{(n)}^f = \underbrace{C_1 \left(1 + \frac{3\|\mathbf{v}_{(n)}^{\text{UAV}}\|^2}{U_{\text{tip}}^2} \right)}_{\text{blade profile}} +$$

$$C_2 \underbrace{\sqrt{\sqrt{C_3 + \frac{\|\mathbf{v}_{(n)}^{\text{UAV}}\|^4}{4}} - \frac{\|\mathbf{v}_{(n)}^{\text{UAV}}\|^2}{2}}}_{\text{induced}} + \underbrace{C_4 \|\mathbf{v}_{(n)}^{\text{UAV}}\|^3}_{\text{parasite}}, \quad (11)$$

where U_{tip} refers to the rotor's tip speed, and C_1, C_2, C_3 , and C_4 are constants related to the UAV's weight and its aerodynamic parameters [7]. As such, the flight energy consumption at time slot n is given by

$$E_{(n)}^f = P_{(n)}^f \delta. \quad (12)$$

Denote $\bar{E}^f$ the mean flight energy consumption during the flight, which is calculated as

$$\bar{E}^f = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} E_{(n)}^f. \quad (13)$$

Given the UAV's limited energy budget, flight energy consumption must remain within acceptable limits. The constraint on flight energy consumption is

$$\bar{E}^f \leq \bar{E}_{\max}^f. \quad (14)$$

Here, $\bar{E}_{\max}^f$ represents the maximum permissible mean energy consumption. Constraint (14) ensures that the UAV operates within its energy budget, which allows for efficient and sustainable flight operations.

D. Problem Formulation

To mitigate the significant uplink interference caused by the UAV to non-associated BSs and to balance the trade-off between A2G and G2G uplink throughput. We consider the weighted throughput at time slot n , which is given by:

$$Z(\{\mathbf{s}_{(n)}^{\text{UAV}}, j(m, n), p(m, n)\}) \triangleq \lambda_u \sum_{m \in \mathcal{M}} \delta R_u(m, n) + \lambda_g \sum_{m \in \mathcal{M}} \delta R_g(m, n). \quad (15)$$

Here, $\lambda_u \geq 0$ and $\lambda_g \geq 0$ represent constant weights allocated to the A2G and G2G throughput, respectively. Due to the randomness introduced by the target's movements, our objective is to maximize the long-term average weighted throughput by optimizing the UAV's trajectory $\{\mathbf{s}_{(n)}^{\text{UAV}}\}$, the uplink BS associations $\{j(m, n)\}$, and UAV's transmit power allocation $\{p(m, n)\}$. This problem is formulated as a multi-stage stochastic optimization problem:

$$(P1) : \max_{\{\mathbf{s}_{(n)}^{\text{UAV}}, j(m, n), p(m, n)\}} \lim_{N \rightarrow +\infty} \frac{\sum_{n \in \mathcal{N}} Z(\{\mathbf{s}_{(n)}^{\text{UAV}}, j(m, n), p(m, n)\})}{N}$$

s.t. (1), (6), (14),

$$j(m, n) \in \mathcal{J}_{m,n}^c, p(m, n) \geq 0, \quad \forall m \in \mathcal{M}, n \in \mathcal{N}, \quad (16)$$

$$\sum_{m \in \mathcal{M}} p(m, n) \leq P_{\max}, \quad \forall n \in \mathcal{N}, \quad (17)$$

where $P_{\max}$ represents the maximum transmit power of the UAV. Constraint (6) ensures that the UAV keeps the moving target within its reliable tracking range D . Constraint (16)

ensures that the UAV can associate with BS $j(m, n)$ during time slot n only if the RB m is not occupied by any GD. By enforcing this constraint, we prevent potential collisions and ensure efficient use of communication resources between the UAV and the BSs. Constraint (17) represents the maximum transmit power $P_{\max}$ constraint for the UAV. This constraint directly influences communication energy consumption, which balances communication performance with energy efficiency, thus influencing the UAV's operational behavior and network performance.

Problem (P1) is inherently non-convex and characterized by the randomness of the target's movements. Such variability necessitates sophisticated management to ensure that the UAV operates efficiently within its limited energy budget. Solving (P1) requires knowledge of energy consumption across all N time slots, which is impractical to predict in advance. To address this challenge, we develop an optimization framework based on Lyapunov theory [44] that only utilizes energy information from past and current time slots. In the following, Section IV introduces the theoretical framework necessary for transforming and tackling the multi-stage stochastic optimization problem to manageable deterministic problems for each time slot, Section V details the practical implementation of the optimization within each time slot.

IV. PROPOSED LYAPUNOV-BASED OPTIMIZATION FRAMEWORK

In this section, we develop a Lyapunov-based optimization framework to solve the problem (P1). This framework is particularly effective in addressing the uncertainties and dynamics inherent in stochastic optimization problems by focusing on long-term stability. First, we convert the multi-stage stochastic problem into a single time-slot deterministic problem using Lyapunov optimization theory. To achieve this, we establish a virtual energy-sensitive queue $\{Q[n]\}$ for the constraint (14), which is designed to track and manage the UAV's energy consumption over time. This queue evolves as follows:

$$Q[n+1] = \max \left\{ Q[n] - \bar{E}_{\max}^f, 0 \right\} + E_{(n)}^f, \quad (18)$$

where $Q[n]$ represents the cumulative energy imbalance and $Q[0] = 0$. In (18), the energy consumption $E_{(n)}^f$ acts as the arrival rate of $Q[n]$, while $\bar{E}_{\max}^f$ acts as the service rate of $Q[n]$. By maintaining the stability of $Q[n]$, the Lyapunov framework ensures that the system can adapt to stochastic variations in energy consumption. The evolution of this queue helps in monitoring whether the UAV's energy usage is within acceptable limits. According to [44], queue stability is defined as follows:

Definition 1: A discrete-time queue $\{Q[n]\}$ achieves strong stability if:

$$\limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \mathbb{E}[Q[n]] \leq +\infty. \quad (19)$$

Based on this definition, we provide the following lemma.

Lemma 1: Stability of the virtual queue $\{Q[n]\}$ ensures that constraint (14) is met.

Proof: From the evolution of $\{Q[n]\}$, we have

$$Q[n+1] \geq Q[n] - \bar{E}_{\max}^f + E_{(n)}^f, \forall n. \quad (20)$$

Taking the sum of both sides of (20) and dividing it by N , we obtain:

$$\frac{Q[N] - Q[0]}{N} + \bar{E}_{\max}^f \geq \frac{1}{N} \sum_{n=0}^{N-1} E_{(n)}^f. \quad (21)$$

According to Definition 1, if the virtual queue $Q[n]$ is stable, then we obtain $\lim_{N \rightarrow \infty} (\mathbb{E}[Q[N]]/N) = 0$. As $N \rightarrow \infty$, $\frac{Q[N]}{N}$ approaches zero if $Q[n]$ is stable, which confirms that the constraint (14) is satisfied due to (21). $\square$

We define a quadratic Lyapunov function to measure queue congestion as

$$J(Q[n]) = \frac{1}{2}(Q[n])^2. \quad (22)$$

This quadratic form is chosen for its ability to effectively capture queue congestion severity by imposing higher penalties on larger queue states, ensuring mathematical tractability and facilitating the derivation of stability guarantees [7], [8]. In particular, a lower $J(Q[n])$ suggests that $Q[n]$ remains small, thereby guaranteeing the stability of $Q[n]$. To ensure stability over all N time slots, $J(Q[n])$ must be kept small in each slot. To achieve this, we define the conditional Lyapunov drift function, which measures the change in $J(Q[n])$ between two consecutive time slots, i.e.,

$$\Delta(Q[n]) = \mathbb{E}[J(Q[n+1]) - J(Q[n]) | Q[n]]. \quad (23)$$

Following (23), we establish the drift-plus-penalty function as

$$\begin{aligned} \Delta_V(Q[n]) &= \Delta(Q[n]) \\ &\quad - V \mathbb{E}[Z(\{s_{(n)}^{\text{UAV}}, j(m, n), p(m, n)\}) | Q[n]]. \end{aligned} \quad (24)$$

Here, $V \geq 0$ represents the Lyapunov parameter, balancing the tradeoff between the weighted network throughput and the Lyapunov drift function. This balance is crucial for optimizing performance while maintaining system stability under stochastic conditions. Thus, optimizing trajectory and resource allocation for maximum weighted network throughput and queue stability is achieved by minimizing the drift-plus-penalty function at each time slot. Notably, the conditional Lyapunov drift function defined in (23) is relatively complex to compute. To tackle such issue, we first derive its upper bound by using the following Theorem 1.

Theorem 1: Assume that $\{s_{(n)}^{\text{UAV}}, j(m, n), p(m, n)\}$ is a feasible solution to problem (P1), we can obtain an upper bound for the drift-plus-penalty function as follows:

$$\begin{aligned} \Delta_V(Q[n]) &\leq D_n - \mathbb{E} \left[Q[n] \left(\bar{E}_{\max}^f - E_{(n)}^f \right) | Q[n] \right] \\ &\quad - V \mathbb{E} \{ Z(\{s_{(n)}^{\text{UAV}}, j(m, n), p(m, n)\}) | Q[n] \}, \end{aligned} \quad (25)$$

where D_n is a constant.

Proof: By squaring the update rule for $Q[n]$ in (18), we obtain

$$(Q[n+1])^2 = \left(\max \left\{ Q[n] - \bar{E}_{\max}^f, 0 \right\} + E_{(n)}^f \right)^2. \quad (26)$$

By using the inequality $(\max\{a - b, 0\} + c)^2 \leq a^2 + b^2 + c^2 + 2a(c - b)$, we derive

$$(Q[n+1])^2 \leq (Q[n])^2 + \left(\bar{E}_{\max}^f\right)^2 + \left(E_{(n)}^f\right)^2 + 2Q[n]\left(E_{(n)}^f - \bar{E}_{\max}^f\right). \quad (27)$$

Based on (27) and the definition provided in (22), we obtain

$$\begin{aligned} J(Q[n+1]) - J(Q[n]) &= \frac{1}{2}(Q[n+1])^2 - \frac{1}{2}(Q[n])^2 \\ &\leq D_n + Q[n]\left(E_{(n)}^f - \bar{E}_{\max}^f\right), \end{aligned} \quad (28)$$

where D_n is a constant, given by:

$$D_n = \frac{1}{2}\left(\bar{E}_{\max}^f\right)^2 + \frac{1}{2}\left(E_{\max}^f\right)^2, \quad (29)$$

where $E_{\max}^f$ represents the maximum flight energy consumption of a UAV across all time slots. Therefore, the upper bound for the conditional Lyapunov drift function is:

$$\Delta(Q[n]) \leq D_n - \mathbb{E}\left[Q[n]\left(\bar{E}_{\max}^f - E_{(n)}^f\right) \middle| Q[n]\right]. \quad (30)$$

The upper bound derived in (30) can be employed to derive the upper bound expression in (25) and the proof concludes. $\square$

Inspired by Theorem 1, we minimize the upper bound from (25) rather than directly minimizing the drift-plus-penalty function at each time slot. By removing the constant term D_n , we minimize the following term at each time slot n , i.e.,

$$\begin{aligned} F(\{s_{(n)}^{\text{UAV}}, j(m, n), p(m, n)\}) &\triangleq Q[n]E_{(n)}^f \\ &- VZ(\{s_{(n)}^{\text{UAV}}, j(m, n), p(m, n)\}). \end{aligned} \quad (31)$$

As a result, solving problem (P1) can be transformed to solving the following problem (P2) for each time slot:

$$\begin{aligned} \text{(P2): } &\min_{\{s_{(n)}^{\text{UAV}}, j(m, n), p(m, n)\}} F(\{s_{(n)}^{\text{UAV}}, j(m, n), p(m, n)\}) \\ \text{s.t. } &\text{(1), (6), (16), (17)}. \end{aligned}$$

The overall online control algorithm for solving (P1) is summarized in Algorithm 1. To illustrate the overall workflow of our solution, Algorithm 1 operates over discrete time slots to manage the UAV's operations efficiently. At each time slot n , the algorithm retrieves the current state of the virtual energy queue $Q[n]$, solves the deterministic optimization problem (P2) to determine the optimal UAV trajectory, BS associations, and power allocations, and then updates $Q[n]$ based on the UAV's energy consumption. This process ensures online adaptability, maintains energy constraints, and minimizes interference. The Lyapunov framework plays a crucial role in enforcing the energy constraint. By updating the virtual energy queue $Q[n]$ based on the UAV's energy consumption during each time slot, the framework ensures that the long-term average energy consumption remains within the permissible limits defined by constraint (14), as shown in Lemma 1.

Algorithm 1: Lyapunov-Based Online Control Algorithm for Problem (P1).

```

1: while  $n \leq N$  do
2:   Obtain  $Q[n]$ ;
3:   Solve problem (P2) to obtain optimized solution
       $\{s_{(n)}^{\text{UAV}}, j(m, n), p(m, n)\}$ ;
4:   Update the queue  $Q[n]$  according to (18);
5:   Update  $n = n + 1$ ;
6: end while

```

V. SINGLE TIME SLOT OPTIMIZATION

In this section, we focus on solving the deterministic per-slot optimization problem (P2) at time slot n . Though problem (P2) remains non-convex, it can be split into two subproblems and solved iteratively using alternating optimization technique until convergence. Specifically, the two subproblems are: 1) The UAV's trajectory optimization subproblem, 2) The BS association and transmit power allocation subproblem. In the following, we will explain how each subproblem is addressed.

A. UAV's Trajectory Optimization Subproblem

Given the transmission power allocation $\{p(m, n)\}$ and BS association $\{j(m, n)\}$, the UAV trajectory design subproblem is written as

$$\begin{aligned} \text{(P3): } &\min_{\{s_{(n)}^{\text{UAV}}, \mathbf{v}_{(n)}^{\text{UAV}}\}} Q[n]\delta \left(C_1 \left(1 + \frac{3\|\mathbf{v}_{(n)}^{\text{UAV}}\|^2}{U_{\text{tip}}^2} \right) + \right. \\ &C_2 \left(\sqrt{C_3 + \frac{\|\mathbf{v}_{(n)}^{\text{UAV}}\|^4}{4}} - \frac{\|\mathbf{v}_{(n)}^{\text{UAV}}\|^2}{2} \right)^{1/2} + C_4 \|\mathbf{v}_{(n)}^{\text{UAV}}\|^3 \Big) \\ &- V \sum_{m \in \mathcal{M}} \delta(\lambda_u R_u(m, n) + \lambda_g R_g(m, n)) \\ \text{s.t. } &\text{(1), (6),} \\ &\mathbf{v}_{(n)}^{\text{UAV}} = (\mathbf{s}_{(n)}^{\text{UAV}} - \mathbf{s}_{(n-1)}^{\text{UAV}})/\delta, \quad \forall n. \end{aligned} \quad (32)$$

The objective function in (P3) is non-convex with respect to $\mathbf{s}_{(n)}^{\text{UAV}}$ due to the non-convex terms $TM_{(n)} \triangleq \sqrt{\left(\sqrt{C_3 + \frac{\|\mathbf{v}_{(n)}^{\text{UAV}}\|^4}{4}} - \frac{\|\mathbf{v}_{(n)}^{\text{UAV}}\|^2}{2} \right)}$ and $TM_{(m, n)} \triangleq -(\lambda_u R_u(m, n) + \lambda_g R_g(m, n))$, $m \in \mathcal{M}$, $n \in \mathcal{N}$. We first transform the objective function into a convex function by introducing slack variables. Specifically, for the non-convex term $TM_{(n)}$, we introduce the slack variable $\tau_{(n)}$ such that:

$$\tau_{(n)} \geq \sqrt{\left(\sqrt{C_3 + \frac{\|\mathbf{v}_{(n)}^{\text{UAV}}\|^4}{4}} - \frac{\|\mathbf{v}_{(n)}^{\text{UAV}}\|^2}{2} \right)},$$

then we have

$$\frac{C_3}{\tau_{(n)}^2} \leq \tau_{(n)}^2 + \|\mathbf{v}_{(n)}^{\text{UAV}}\|^2. \quad (33)$$

To handle the non-convex term $TM_{(m,n)}$, we introduce slack variables $y_{(m,n)}$ and $z_{(m,n)}$, and have the following constraints:

$$y_{(m,n)} \geq \|\mathbf{s}_{(n)}^{\text{UAV}} - \mathbf{s}_{j(m,n)}^{\text{BS}}\|^2, \quad \forall m, n, \quad (34)$$

$$z_{(m,n)} \geq (H^2 + \|\mathbf{s}_{(n)}^{\text{UAV}} - \mathbf{s}_{j(m,n)}^{\text{BS}}\|^2)^{-1}, \quad \forall m, n. \quad (35)$$

With such relaxation, problem (P3) can be transformed into

$$\begin{aligned} \text{(P4)} : \quad & \min_{\{\mathbf{s}_{(n)}^{\text{UAV}}, \mathbf{v}_{(n)}^{\text{UAV}}, \tau_{(n)}, y_{(m,n)}, z_{(m,n)}\}} Q[\mathbf{n}]\delta \left(C_1 + \frac{3C_1 \|\mathbf{v}_{(n)}^{\text{UAV}}\|^2}{U_{\text{up}}^2} \right. \\ & \left. + C_4 \|\mathbf{v}_{(n)}^{\text{UAV}}\|^3 + C_2 \tau_{(n)} \right) - V\delta \sum_{m \in \mathcal{M}} B \\ & \times \left[\lambda_u \log_2 \left(1 + \frac{p(m,n)\beta_0}{\sigma_{j(m,n),m,n}^2 \times (H^2 + y_{(m,n)})^{\alpha/2}} \right) \right. \\ & \left. + \lambda_g \sum_{j \in \mathcal{J}_{m,n}} \log_2 \left(1 + \frac{\hat{p}_{j,m,n} H_{j,m,n}}{\sigma_{j,m,n}^2 + p(m,n)\beta_0 z_{(m,n)}^{\alpha/2}} \right) \right] \end{aligned}$$

s.t. (1), (6), (32)–(35).

Theorem 2: Problem (P4) is equivalent to problem (P3).

Proof: Suppose $\{\mathbf{s}_{(n)}^{\text{UAV},*}, \tau_{(n)}^*, y_{(m,n)}^*, z_{(m,n)}^*\}$ is the optimal solution of problem (P4), then the following equation holds:

$$\begin{aligned} \tau_{(n)}^* &= \sqrt{\left(\sqrt{C_3 + \frac{\|\mathbf{v}_{(n)}^{\text{UAV},*}\|^4}{4}} - \frac{\|\mathbf{v}_{(n)}^{\text{UAV},*}\|^2}{2} \right)}, \quad \forall n, \\ y_{(m,n)}^* &= \|\mathbf{s}_{(n)}^{\text{UAV},*} - \mathbf{s}_{j(m,n)}^{\text{BS}}\|^2, \quad \forall m, n, \\ z_{(m,n)}^* &= (H^2 + \|\mathbf{s}_{(n)}^{\text{UAV},*} - \mathbf{s}_{j(m,n)}^{\text{BS}}\|^2)^{-1}, \quad \forall m, n, \end{aligned} \quad (36)$$

where $\mathbf{v}_{(n)}^{\text{UAV},*} = (\mathbf{s}_{(n)}^{\text{UAV},*} - \mathbf{s}_{(n-1)}^{\text{UAV}})/\delta$. Since otherwise, we can further reduce the objective value by choosing a smaller $\tau_{(n)}$, $y_{(m,n)}$ and $z_{(m,n)}$ without violating the constraints (33)–(35). Therefore, problem (P4) is equivalent to the problem (P3) due to (36). $\square$

For problem (P4), the objective function and the constraints (33) and (35) remain non-convex. Similar to [7], the SCA method is adopted to tackle the non-convexity. The central concept of SCA involves approximating the non-convex problem as a convex one at each local point in every iteration. We achieve a locally optimal solution for (P4) through iterative resolution of a series of approximated convex optimization problems.

Proposition 1: For any vector $\mathbf{x} \in \mathbb{R}^n$ with a given point $\mathbf{x}^l \in \mathbb{R}^n$, the norm squared $\|\mathbf{x}\|^2$ is lower-bounded by

$$\|\mathbf{x}\|^2 \geq \|\mathbf{x}^l\|^2 + 2(\mathbf{x}^l)^\top (\mathbf{x} - \mathbf{x}^l), \quad (37)$$

where $[\cdot]^\top$ denotes the transpose.

Proof: Consider the function $g(\mathbf{x}) = \|\mathbf{x}\|^2$, which is a convex function. The first-order Taylor expansion of g at $\mathbf{x}^l$ is given by

$$g(\mathbf{x}) \geq g(\mathbf{x}^l) + \nabla g(\mathbf{x}^l)^\top (\mathbf{x} - \mathbf{x}^l), \quad (38)$$

where $\nabla g(\mathbf{x}^l) = 2\mathbf{x}^l$. Substituting $g(\mathbf{x}) = \|\mathbf{x}\|^2$ and $g(\mathbf{x}^l) = \|\mathbf{x}^l\|^2$, we obtain the conclusion. $\square$

Given the function $f(\mathbf{s}_{(n)}^{\text{UAV}}, \tau_{(n)}) \triangleq \tau_{(n)}^2 + \|\mathbf{v}_{(n)}^{\text{UAV}}\|^2$, and a local point $\mathbf{s}_{(n)}^{\text{UAV},l}$ at the l th iteration, we can obtain the lower bound $f^l(\mathbf{s}_{(n)}^{\text{UAV}}, \tau_{(n)})$ by applying Proposition 1 as: $f^l(\mathbf{s}_{(n)}^{\text{UAV}}, \tau_{(n)}) \triangleq (\tau_{(n)}^l)^2 + 2\tau_{(n)}^l(\tau_{(n)} - \tau_{(n)}^l) + \frac{\|\mathbf{s}_{(n)}^{\text{UAV},l} - \mathbf{s}_{(n-1)}^{\text{UAV}}\|^2}{\delta^2} + \frac{2}{\delta^2}(\mathbf{s}_{(n)}^{\text{UAV},l} - \mathbf{s}_{(n-1)}^{\text{UAV}})^\top (\mathbf{s}_{(n)}^{\text{UAV}} - \mathbf{s}_{(n)}^{\text{UAV},l})$, where $\tau_{(n)}^l$ is defined as: $\tau_{(n)}^l = \sqrt{\left(\sqrt{C_3 + \frac{\|\mathbf{s}_{(n)}^{\text{UAV},l} - \mathbf{s}_{(n-1)}^{\text{UAV}}\|^4}{4\delta^4}} - \frac{\|\mathbf{s}_{(n)}^{\text{UAV},l} - \mathbf{s}_{(n-1)}^{\text{UAV}}\|^2}{2\delta^2} \right)}$. Similarly, for

the function $\phi(\mathbf{s}_{(n)}^{\text{UAV}}) \triangleq H^2 + \|\mathbf{s}_{(n)}^{\text{UAV}} - \mathbf{s}_{j(m,n)}^{\text{BS}}\|^2$ and a local point $\mathbf{s}_{(n)}^{\text{UAV},l}$ at the l th iteration, we obtain the lower bound by applying Proposition 1 as: $\phi^l(\mathbf{s}_{(n)}^{\text{UAV}}) \triangleq H^2 + \|\mathbf{s}_{(n)}^{\text{UAV},l} - \mathbf{s}_{j(m,n)}^{\text{BS}}\|^2 + 2(\mathbf{s}_{(n)}^{\text{UAV},l} - \mathbf{s}_{j(m,n)}^{\text{BS}})^\top (\mathbf{s}_{(n)}^{\text{UAV}} - \mathbf{s}_{(n)}^{\text{UAV},l})$, $\forall m, n$.

Based on the above analysis, at the l th iteration, constraints (33) and (35) can be approximated as:

$$\frac{C_3}{\tau_{(n)}^2} \leq f^l(\mathbf{s}_{(n)}^{\text{UAV}}, \tau_{(n)}), \quad \forall n, \quad (39)$$

$$\frac{1}{z_{(m,n)}} \leq \phi^l(\mathbf{s}_{(n)}^{\text{UAV}}), \quad \forall m, n, \quad (40)$$

where $f^l(\mathbf{s}_{(n)}^{\text{UAV}}, \tau_{(n)})$ and $\phi^l(\mathbf{s}_{(n)}^{\text{UAV}})$ are linear functions, thus ensuring (39) and (40) are convex constraints.

Proposition 2: For any variable x with a given point x^l and $\alpha \geq 2$, the function $\log_2(1 + \frac{A}{C + x^{\alpha/2}})$ is lower-bounded as follows:

$$\begin{aligned} \log_2 \left(1 + \frac{A}{C + x^{\alpha/2}} \right) &\geq \log_2 \left(1 + \frac{A}{C + (x^l)^{\alpha/2}} \right) \\ &+ \frac{A \cdot \frac{\alpha}{2} (x^l)^{\alpha/2-1}}{\ln(2) (C + (x^l)^{\alpha/2}) \left(1 + \frac{A}{C + (x^l)^{\alpha/2}} \right)} (x - x^l). \end{aligned} \quad (41)$$

Proof: Consider the function $f(x) = \log_2(1 + \frac{A}{C + x^{\alpha/2}})$. Due to (38), we compute the derivative $f'(x) = \frac{1}{\ln(2)} \cdot \frac{-A \cdot \frac{\alpha}{2} x^{\alpha/2-1}}{(C + x^{\alpha/2})(1 + \frac{A}{C + x^{\alpha/2}})}$. Then, substituting $f(x^l)$ and $f'(x^l)$ into (38), we obtain the result in (41). $\square$

Given functions $g_1(y_{(m,n)}) \triangleq \lambda_u \log_2(1 + \frac{p(m,n)\beta_0}{\sigma_{j(m,n),m,n}^2 (H^2 + y_{(m,n)})^{\alpha/2}})$ and $g_2(z_{(m,n)}) \triangleq \lambda_g \sum_{j \in \mathcal{J}_{m,n}} \log_2(1 + \frac{\hat{p}_{j,m,n} H_{j,m,n}}{\sigma_{j,m,n}^2 + p(m,n)\beta_0 z_{(m,n)}^{\alpha/2}})$. By using Proposition 2, we can approximate g_1 and g_2 to obtain the desired lower bounds, i.e.,

$$\begin{aligned} g_1^l(y_{(m,n)}) &\triangleq \lambda_u \log_2 \left(1 + \frac{p(m,n)\beta_0}{\sigma_{j(m,n),m,n}^2 (H^2 + (y_{(m,n)}^l)^{\alpha/2})} \right) \\ &- \lambda_u \cdot \frac{\alpha}{2 \ln(2)} \cdot \frac{p(m,n)\beta_0}{\sigma_{j(m,n),m,n}^2} \cdot \frac{(y_{(m,n)} - y_{(m,n)}^l)}{(H^2 + (y_{(m,n)}^l)^{\alpha/2+1})} \\ &\times \frac{1}{1 + \frac{p(m,n)\beta_0}{\sigma_{j(m,n),m,n}^2 (H^2 + (y_{(m,n)}^l)^{\alpha/2})}}, \end{aligned}$$

Algorithm 2: Iterative Algorithm to UAV's Trajectory Optimization.

- 1: Initialize $\{s_{(n)}^{\text{UAV},1}, S^1\}$;
 - 2: Set $l \leftarrow 1$;
 - 3: **repeat**
 - 4: Obtain the optimal solution for (P5) as $\{s_{(n)}^{\text{UAV},l+1}, S^{l+1}\}$ based on $\{s_{(n)}^{\text{UAV},l}\}$;
 - 5: Update $l \leftarrow l + 1$;
 - 6: **until** $\frac{|S^{l+1} - S^l|}{S^l} \leq \epsilon$, where ϵ denotes the accuracy.
-

and $g_2^l(z_{(m,n)}) \triangleq$

$$\lambda_g \sum_{j \in \mathcal{J}_{m,n}} \left(\log_2 \left(1 + \frac{\hat{p}_{j,m,n} H_{j,m,n}}{\sigma_{j,m,n}^2 + p(m,n) \beta_0 (z_{(m,n)}^l)^{\alpha/2}} \right) - \frac{\alpha \cdot \hat{p}_{j,m,n} H_{j,m,n} \cdot (z_{(m,n)} - z_{(m,n)}^l)}{2 \ln(2) (\sigma_{j,m,n}^2 + p(m,n) \beta_0 (z_{(m,n)}^l)^{\alpha/2})} \times \frac{p(m,n) \beta_0 (z_{(m,n)}^l)^{\alpha/2-1}}{(\sigma_{j,m,n}^2 + p(m,n) \beta_0 (z_{(m,n)}^l)^{\alpha/2} + \hat{p}_{j,m,n} H_{j,m,n})} \right).$$

Therefore, problem (P4) is approximated as

$$\begin{aligned} \text{(P5)} : \quad & \min_{\{s_{(n)}^{\text{UAV}}, \tau_{(n)}, y_{(m,n)}, z_{(m,n)}\}} Q[n] \delta \left(C_1 + \frac{3C_1 \|\mathbf{v}_{(n)}^{\text{UAV}}\|^2}{U_{\text{tip}}^2} \right. \\ & \left. + C_4 \|\mathbf{v}_{(n)}^{\text{UAV}}\|^3 + C_2 \tau_{(n)} \right) \\ & - VB \delta \sum_{m \in \mathcal{M}} \left(g_1^l(y_{(m,n)}) + g_2^l(z_{(m,n)}) \right) \end{aligned}$$

s.t. (1), (6), (32), (34), (39), (40).

Problem (P5) is convex and standard convex optimization solvers such as CVX [45] can be utilized to solve (P5) efficiently. Note that due to the global lower bounds established in (39) and (40) and Proposition 2, satisfying the constraints of problem (P5) ensures that the constraints of the original problem (P4) are also met. By iteratively updating the local points at each iteration through solving (P5) with SCA technique, we develop an efficient algorithm for addressing the non-convex problem (P4) or its original form (P3), as outlined in Algorithm 2, where S^l represent the objective value for (P5) in the l th iteration. By following similar arguments in [43], [46], it can be demonstrated that Algorithm 2 with SCA technique is guaranteed to converge to a solution that satisfies the Karush-Kuhn-Tucker (KKT) conditions of problem (P4).

B. The BS Association and Transmit Power Allocation Subproblem

For given UAV's trajectory $\{s_{(n)}^{\text{UAV}}\}$, the BS association and transmit power allocation subproblem can be expressed as

$$\text{(P6)} : \quad \min_{\{j(m,n), p(m,n)\}} -V \left(\lambda_u \sum_{m \in \mathcal{M}} \delta R_u(m,n) + \lambda_g \sum_{m \in \mathcal{M}} \delta R_g(m,n) \right)$$

s.t. (16), (17).

From the following Theorem, we can easily obtain the optimal BS association for (P6).

Theorem 3: The optimal BS association to problem (P6) can be obtained as $j^*(m,n) = \arg \max_{j \in \mathcal{J}_{m,n}} \frac{G_n^{j(m,n)}}{\sigma_{j(m,n),m,n}^2}, \forall m \in \mathcal{M}$.

Proof: We prove it by contradiction. Assume that $j^*(m,n)$ is not the optimal BS association. Note that the BS association $\{j(m,n)\}$ is only related to A2G uplink transmission rate $R_u(m,n)$. This assumption implies that there exists another feasible cell association solution, denoted by $j^0(m,n)$, such that the A2G throughput with $j^0(m,n)$ is higher than that with $j^*(m,n)$. Thus, we have

$$\begin{aligned} \sum_{m \in \mathcal{M}} \log_2 \left(1 + \frac{p(m,n) G_n^{j^0(m,n)}}{\sigma_{j^0(m,n),m,n}^2} \right) &> \\ \sum_{m \in \mathcal{M}} \log_2 \left(1 + \frac{p(m,n) G_n^{j^*(m,n)}}{\sigma_{j^*(m,n),m,n}^2} \right). \end{aligned} \quad (42)$$

However, due to the definition of $j^*(m,n)$, we have $\frac{G_n^{j^*(m,n)}}{\sigma_{j^*(m,n),m,n}^2} \geq \frac{G_n^{j^0(m,n)}}{\sigma_{j^0(m,n),m,n}^2}, \forall m \in \mathcal{M}$. Since $\log_2(x)$ is a monotonically increasing function, this implies that $\sum_{m \in \mathcal{M}} \log_2(1 + \frac{p(m,n) G_n^{j^*(m,n)}}{\sigma_{j^*(m,n),m,n}^2}) \geq \sum_{m \in \mathcal{M}} \log_2(1 + \frac{p(m,n) G_n^{j^0(m,n)}}{\sigma_{j^0(m,n),m,n}^2})$. This directly contradicts (42) that the A2G throughput with $j^0(m,n)$ is higher than that with $j^*(m,n)$. Therefore, $j^*(m,n)$ is the optimal BS association to problem (P6), which concludes the proof. $\square$

Theorem 3 demonstrates that the optimal BS association remains unaffected by the UAV's transmit power allocation $\{p(m,n)\}$, which simplifies the optimization process. For brevity, we introduce notation $G_{m,n}^* \triangleq \frac{G_n^{j^*(m,n)}}{\sigma_{j^*(m,n),m,n}^2}, \forall m \in \mathcal{M}, n \in \mathcal{N}$, and (P6) is simplified as

$$\begin{aligned} \text{(P7)} : \quad & \max_{\{p(m,n)\}} \lambda_u \sum_{m \in \mathcal{M}} \delta \log_2(1 + p(m,n) G_{m,n}^*) \\ & + \lambda_g \sum_{m \in \mathcal{M}} \delta R_g(m,n) \\ \text{s.t.} \quad & \sum_{m \in \mathcal{M}} p(m,n) \leq P_{\max}, \forall n \in \mathcal{N}, \quad (43) \\ & p(m,n) \geq 0, \forall m \in \mathcal{M}, n \in \mathcal{N}. \quad (44) \end{aligned}$$

Due to its non-concave objective function, problem (P7) is a non-convex optimization problem. To address this challenge, we employ the SCA technique to achieve a locally optimal solution. In the r th iteration, we introduce $p^r(m,n)$ as the power allocation for the UAV. Given $\{p^r(m,n)\}$, we can obtain a lower bound for $R_g(m,n)$ by employing first-order Taylor approximation, i.e.,

$$R_g(m,n) \geq \sum_{j \in \mathcal{J}_{m,n}} B \log_2 \left(1 + \frac{\hat{p}_{j,m,n} H_{j,m,n}}{\sigma_{j,m,n}^2 E^r(m,n)} \right)$$

Algorithm 3: Iterative Algorithm to Transmit Power Allocation.

- 1: Initialize $\{p^1(m, n), P^1\}$;
- 2: Set $r \leftarrow 1$;
- 3: **repeat**
- 4: Obtain the optimal solution for (P8) as $\{p^{r+1}(m, n), P^{r+1}\}$ based on $\{p^r(m, n)\}$;
- 5: Update $r \leftarrow r + 1$;
- 6: **until** $\frac{|P^{r+1} - P^r|}{P^r} \leq \epsilon$, where ϵ denotes the accuracy.

Algorithm 4: Alternative Optimization Algorithm (AOA) for Problem (P2).

- 1: Initialize $\{s_{(n)}^{\text{UAV},0}, j^0(m, n), p^0(m, n)\}$;
- 2: Set the iteration index and the threshold parameter as $r = 0$ and $\epsilon = 10^{-4}$;
- 3: **repeat**
- 4: With the given $\{p^r(m, n)\}$ and $\{j^r(m, n)\}$, solve the UAV trajectory design subproblem (P3) and obtain the updated UAV position $\{s_{(n)}^{\text{UAV},r+1}\}$ using Algorithm 2;
- 5: With the given $\{s_{(n)}^{\text{UAV},r+1}\}$, obtain the optimal cell association solution $\{j^{r+1}(m, n)\}$ for (P6) according to Theorem 3;
- 6: With the given $\{s_{(n)}^{\text{UAV},r+1}\}$ and $\{j^{r+1}(m, n)\}$, update the transmission power $\{p^{r+1}(m, n)\}$ using Algorithm 3;
- 7: Update the objective value of problem (P2) to F^{r+1} using the updated variables $\{s_{(n)}^{\text{UAV},r+1}\}$, $\{j^{r+1}(m, n)\}$, and $\{p^{r+1}(m, n)\}$;
- 8: Update $r = r + 1$;
- 9: **until** $\frac{|F^{r+1} - F^r|}{F^r} \leq \epsilon$.

$$-C^r(m, n)(p(m, n) - p^r(m, n)), \forall m, n, \quad (45)$$

where $E^r(m, n) = 1 + \frac{p^r(m, n)G_n^j}{\sigma_{j,m,n}^2}$, $C^r(m, n) = \sum_{j \in \mathcal{J}_{m,n}} \frac{G_n^j \hat{p}_{j,m,n} H_{j,m,n}}{(\ln 2)(\sigma_{j,m,n}^2 E^r(m, n) + \hat{p}_{j,m,n} H_{j,m,n}) \sigma_{j,m,n}^2 E^r(m, n)}$. Therefore, in the r th iteration, (P7) is approximated as the following problem, i.e.,

$$\begin{aligned}
 \text{(P8)} : \quad & \max_{\{p(m,n)\}} \lambda_u \delta \sum_{m \in \mathcal{M}} B \log_2(1 + p(m, n) G_{m,n}^u) \\
 & - \lambda_g \delta \sum_{m \in \mathcal{M}} C^r(m, n) p(m, n) \\
 \text{s.t. } & (43), (44),
 \end{aligned}$$

where the objective function omits the constant term for brevity. Note that (P8) is a convex optimization problem and can be solved efficiently by existing optimization tools such as CVX. Let P^r represent the objective value for (P8) in the r th iteration. The following Algorithm 3 summarizes the proposed iterative approach.

To effectively solve problem (P2) at each time slot, we propose an iterative algorithm which optimizes the subproblems (P3)

and (P6) alternatively, and the details are summarized in Algorithm 4. Note that Algorithm 1 simplifies the complex stochastic optimization problem by converting it into a deterministic problem for each time slot. In each time slot, the two subproblems are transformed into convex optimization problems, which are then alternately optimized using Algorithms 2 and 3, both of which employ the interior point method. The computational complexity of this process is primarily influenced by the number of variables [47]. Consequently, the overall computational complexity of Algorithm 4 for each time slot is $\mathcal{O}(LM^{3.5})$, where L denotes the number of iterations with the order $\mathcal{O}(\log(\frac{1}{\epsilon}))$. To further enhance computational efficiency, we note the potential of employing advanced optimization tools such as CVXGEN [48]. CVXGEN is capable of reducing computation times by up to 1000 times compared to standard CVX-based implementations, potentially bringing the running time down to the order of milliseconds. This significant improvement would ensure that the algorithms remain highly efficient and practically affordable for UAV online operations.

VI. SIMULATION RESULTS

In this section, simulations are conducted to evaluate the performance of the proposed schemes for target tracking and communication optimization in cellular-connected UAV networks. The simulations aim to demonstrate the effectiveness of the algorithm in various scenarios, highlighting improvements in network throughput, queue stability, and energy efficiency.

A. Simulation Setup

As shown in Fig. 2, we have set up $J = 61$ BSs, each with a height of 25 meters and an adjacent level set at $q = 2$, with a cell radius of 500 meters. The BSs are distributed across four layers of cellular networks, covering an area with $K = 70$ GDs whose positions are randomly generated. The network operates over $M = 40$ RBs across $T = 300$ s for target tracking tasks, with the UAV operating at an altitude of $H = 100$ m and a time interval of $\delta = 1$ s. For target tracking, the tracking range is set to $D = 300$ m. We consider the practical Rician fading channels [40] with Rician factor K_c , and the CDF function $F(\cdot)$ can be expressed as $F(z) = 1 - Q_1(\sqrt{2K_c}, \sqrt{2(K_c + 1)z})$, where $Q_1(a, b)$ is the Marcum-Q function, with $K_c = 30$ dB, $\Gamma = 5$ dB, $\epsilon = 0.1$. The noise power density is $\sigma^2 = 10^{-14}$ W/Hz with a path loss exponent of $\alpha = 2$, and the channel power gain is $\beta_0 = -50$ dB. Each RB is allocated a bandwidth of $B = 1$ kHz. GDs transmit at a power of $\hat{p}_{j,m,n} = 20$ dBm, while the maximum UAV transmit power is $P_{\max} = 30$ dBm. The UAV's maximum flight speed is $V_{\max} = 50$ m/s, with a maximum average energy consumption of $\bar{E}_{\max}^f = 800$ Joule. Similar to [33], the target's movement follows the Gauss-Markov mobility model, with a mean speed of $\bar{v} = 40$ m/s, a mean direction of $\theta = \pi/4$, and a target randomness level set to $\kappa = 0.8$.

For energy related parameters, we set $C_1 = 7.99$ W and $C_2 = 21.96$ W, as well as aerodynamic constants $C_3 = 263.76$ and $C_4 = 0.0092$. The rotor blade tip speed is $U_{\text{tip}} = 120$ m/s. Unless otherwise stated, the throughput weights are set to $\lambda_u = 1$ and $\lambda_g = 1$, and the penalty weight value is $V = 10$. The parameter

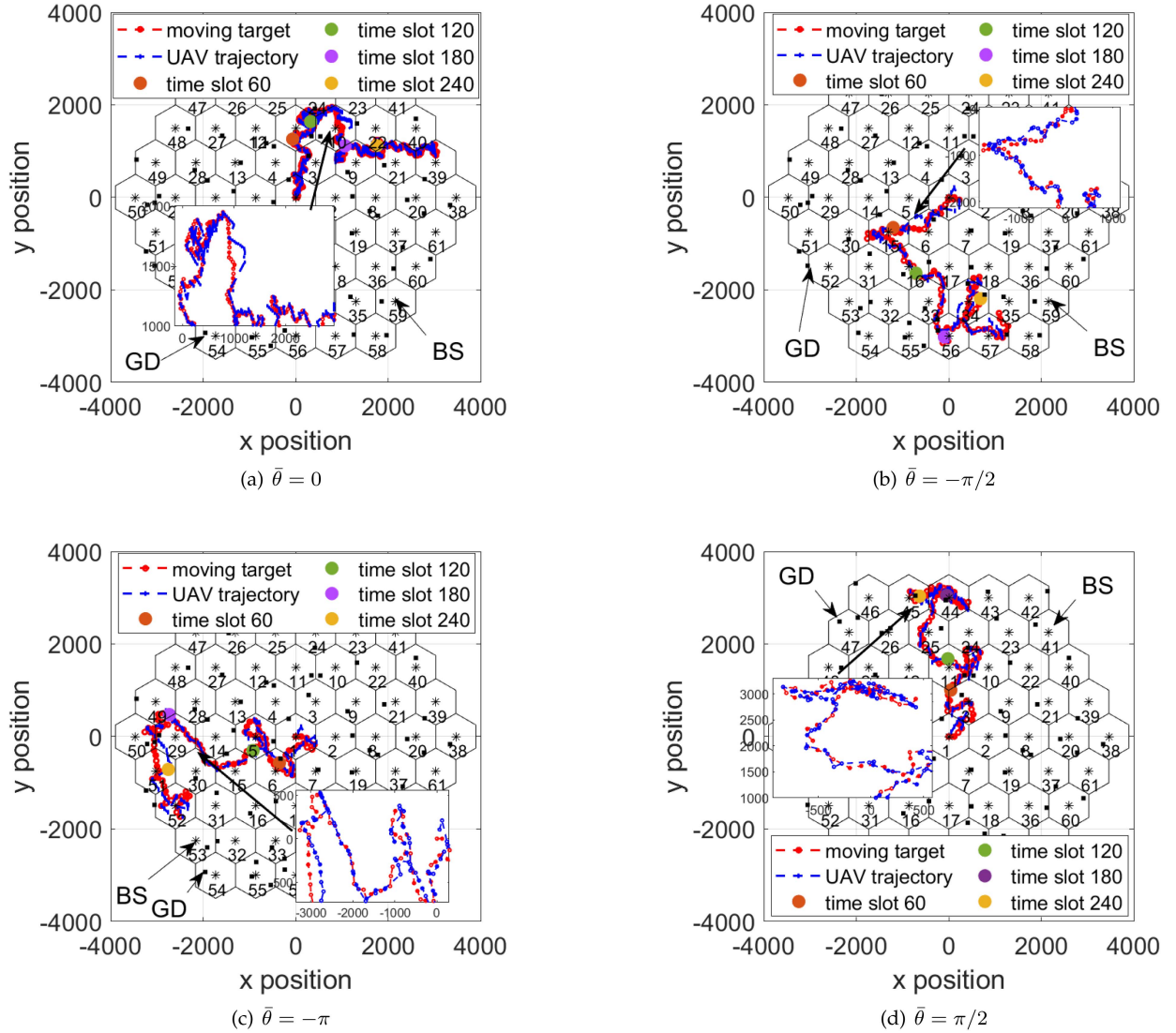


Fig. 2. Optimized trajectory for target tracking.

settings are referenced from relevant literature [3], [4], [30], [32]. All simulations were conducted using MATLAB R2022a with the CVX package and SeDuMi solver on a standard dual-core 3.20 GHz CPU with 32 GB memory.

B. Optimized Tracking Trajectory and Power Allocation With BS Association

Fig. 2 shows the optimized trajectories of the UAV while tracking the target under various movement scenarios. The target's speed $\bar{v}$ is set to 40 m/s, with the direction $\bar{\theta}$ varying among $\{0, -\pi/2, -\pi, \pi/2\}$. These parameters represent different movement patterns of the target, including up, down, left, and right directions. In all scenarios depicted in Fig. 2, the UAV successfully tracks the target, maintaining it within the specified tracking range D . The UAV's trajectory adapts dynamically to the target's movements, ensuring continuous observation. The online adjustment of the UAV's trajectory highlights the

algorithm's robustness and responsiveness, which is crucial for reliable target tracking.

Based on the target movement in Fig. 2(d), Fig. 3 shows the optimized power allocation to different RBs at various time slots, aimed at maximizing throughput within the energy budget. The curves indicate that the UAV dynamically adjusts its power distribution among different RBs to maximize throughput with interference avoidance. Significant variations in power allocation are observed at time slots 60, 120, 180, and 240, with specific RBs receiving higher power to ensure higher throughput with less interference.

Fig. 4 illustrates the BS association under different time slots and RBs in the scenario of Fig. 2(d). Results for other similar scenarios have been omitted for brevity. The UAV's association with various BSs changes over time, which reflects its strategy to connect to the optimal BS with the least interference. For example, at time slot 240, the UAV is associated with BS 20 through RB 18 and RB 29, while the remaining RBs are associated with BS 10. This dynamic association demonstrates the UAV's ability

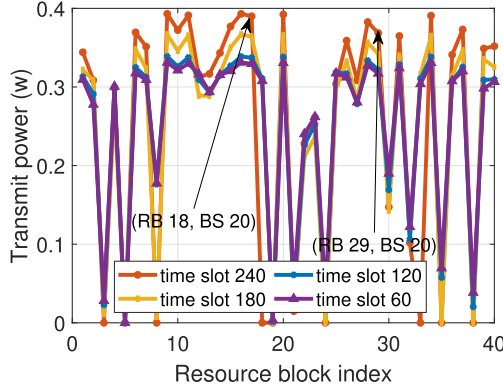


Fig. 3. The power allocation to different RBs at different time slots.

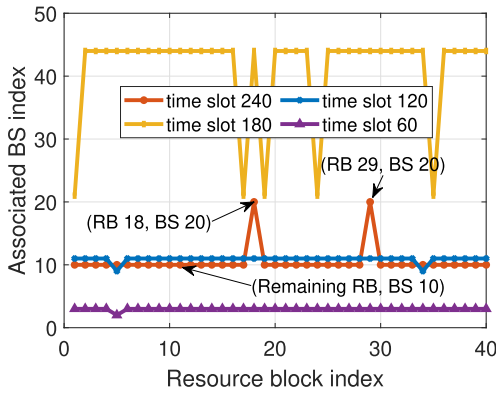


Fig. 4. The BS association under different time slots and RBs.

to select BSs that offer better communication quality, even if they are not the closest in proximity. The UAV dynamically adjusts its trajectory and power allocation in response to the target's movements and the network conditions, demonstrating the robustness and efficiency of the proposed algorithm.

C. Benchmark Comparisons

To better illustrate the superiority of the proposed scheme with AOA, we compare it with several benchmark schemes: 1) *Target path following benchmark (TPF)* [49], where the UAV follows the same trajectory as the target; 2) *High-speed tracking benchmark (HST)*, where the UAV flies at the maximum speed that corresponds to the average maximum flight energy consumption while still satisfying tracking requirements; 3) *Energy-efficient predicted tracking benchmark (EEPT)* [5], where the target location is predicted based on Elman neural network with the aim of energy minimization. 4) *No Lyapunov benchmark (NL)* [27], which strictly requires the energy consumption to meet the target value for each time slot, without utilizing the Lyapunov method. 5) *Nearest BS maximum power benchmark (NBMP)* [50], where the UAV associates with the nearest BS and selects an available RB not occupied by any GDs with maximum transmit power; 6) *Equal power allocation benchmark (EPA)* [51], where the UAV allocates the transmit power equally across all RBs; 7) *Interference-free scheme benchmark (IF)* [30], where the UAV

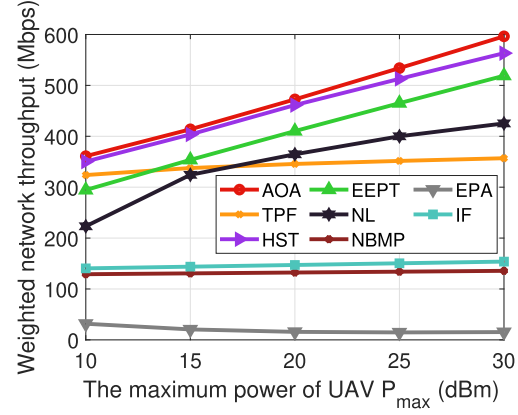


Fig. 5. Performance comparisons between the proposed solution and benchmark schemes.

transmits in the RBs which are not utilized by any GDs in any cells.

Fig. 5 demonstrates the relationship between system throughput and the UAV's maximum transmit power $P_{\max}$. It is observed that the proposed AOA consistently outperforms the benchmark schemes, showing a higher throughput as $P_{\max}$ increases. The TPF benchmark maintains a consistent path, which limits its adaptability and potential communication improvements through UAV trajectory optimization. The HST benchmark ensures maximum speed but fails to balance throughput and energy consumption. The EEPT benchmark prioritizes energy efficiency over communication performance, resulting in lower throughput as it focuses on tracking performance based on predicted tracking, which limits its optimization for communication. The NL benchmark shows relatively lower throughput as it strictly enforces energy consumption limits for each time slot, which results in suboptimal communication performance while ensuring compliance with energy constraints. The NBMP benchmark and EPA benchmark show relatively lower throughput due to inefficient power allocation with more interference. The IF benchmark avoids interference but results in lower throughput due to inefficient use of available RBs. Thus, the proposed solution effectively balances power allocation, interference management, and adaptation to the current communication environment, which results in superior performance in terms of throughput under limited energy budget.

D. Analysis of System Stability and Flight Speed

Fig. 6 shows that while the average queue length (AQL) gradually increases with a higher Lyapunov control parameter V , it remains lower than most of the benchmark schemes, except for the EEPT benchmark, which achieves the smallest AQL by sacrificing throughput. The observed increase in AQL for the proposed AOA as V increases is due to the system prioritizing throughput more as V grows, which inevitably impacts queue stability. In contrast, other benchmarks show constant AQLs because they do not use V to balance throughput and queue length. The EEPT benchmark focuses solely on minimizing energy consumption, leading to the lowest queue length at the

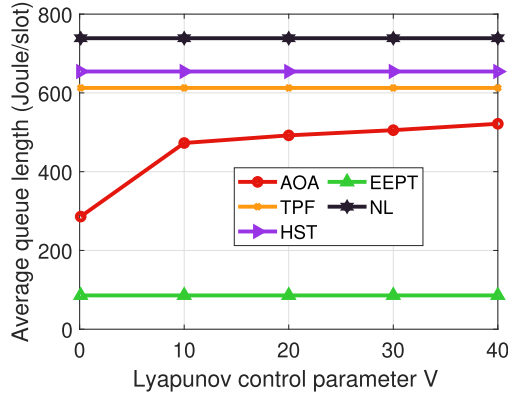


Fig. 6. The average queue length versus control parameter V .

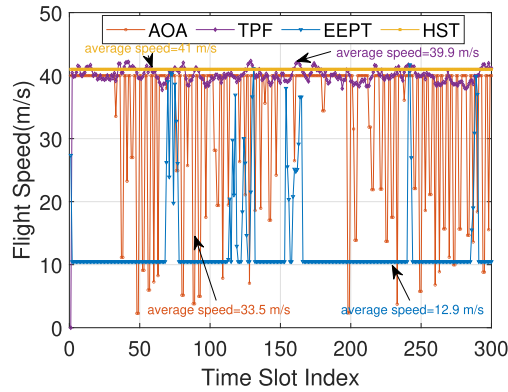


Fig. 7. Optimized UAV's speed under different time slots.

expense of throughput. The proposed *AOA*, while not achieving the minimum AQL, strikes a balance between queue length and throughput, offering a more stable and energy-efficient solution. Other benchmarks, such as the *TPF* benchmark, *HST* benchmark and *NL* benchmark, exhibit larger queue lengths due to their less adaptive nature in handling dynamic communication environments.

Fig. 7 illustrates the UAV's speed during flight for various schemes. Based on the tracking trajectory in Fig. 2(a), our proposed scheme *AOA* dynamically adjusts the UAV's speed in response to the communication environment, maintaining a moderate average speed compared to other benchmarks. This adaptability ensures stable and energy-efficient operation. For the *TPF* benchmark scheme, the UAV's speed closely follows the target's trajectory, resulting in an average speed of 40 m/s. The *HST* benchmark scheme maintains a constant maximum speed of 41 m/s, always moving at its highest possible speed while satisfying energy constraints. The *EEPT* benchmark scheme focuses on minimizing energy usage, resulting in the propulsion power efficient speed of 12.9 m/s. Although it is energy-efficient, it sacrifices throughput performance as previously shown in the performance comparisons. Our proposed scheme achieves an average speed of 33.5 m/s, striking a balance between maximizing throughput and minimizing energy consumption. This online adjustment to environmental conditions contributes to overall system stability and performance.

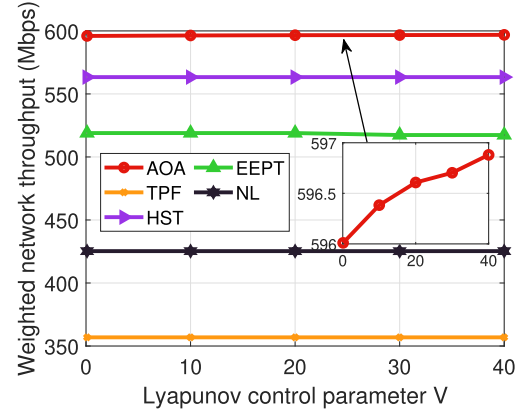


Fig. 8. The throughput versus control parameter V .

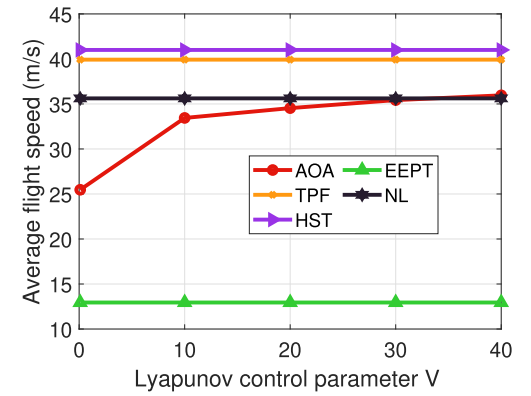


Fig. 9. The average UAV's speed versus control parameter V .

The proposed scheme *AOA* also demonstrates superior performance across various metrics as the Lyapunov control parameter V varies. Fig. 8 shows that it consistently achieves the highest average throughput, highlighting its effectiveness in optimizing communication performance. As the Lyapunov control parameter V increases, the system progressively shifts its focus towards enhancing throughput. This is reflected in the figure, where a noticeable and steady increase in throughput is observed as V becomes larger, indicating the system's capacity to significantly improve throughput as the control parameter is adjusted. In contrast, the *TPF* benchmark achieves the lowest throughput due to its lack of an optimized trajectory planning strategy, which limits its ability to effectively manage communication resources. The *EEPT* benchmark prioritizes minimizing energy consumption through predicted tracking, but this comes at the expense of throughput, resulting in significantly lower throughput values. These benchmarks, by not adequately balancing throughput with their respective goals, fall short in optimizing overall communication performance compared to the proposed scheme. Fig. 9 illustrates the UAV's dynamic speed adjustment, maintaining a moderate average speed compared to other benchmarks, thus balancing throughput and energy efficiency. We can observe that as V increases, the speed of the proposed algorithm also shows an increasing trend, further confirming that with a higher V , the system prioritizes throughput more heavily, leading to increased

energy consumption and impacting queue stability as previously discussed.

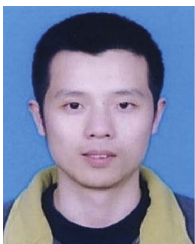
VII. CONCLUSION

This paper explores the challenges of integrating cellular-connected UAV for dynamic target tracking, with a particular focus on managing unpredictable flight energy consumption and mitigating uplink interference. We develop a multi-stage stochastic optimization framework to enhance communication performance by optimizing the long-term average of uplink throughput for both aerial and ground users. By employing the Lyapunov optimization technique, we transformed the complex, stochastic problem into a deterministic format manageable for each time slot. Our proposed online strategy optimizes BS association, power allocation, and UAV trajectory by leveraging the optimal structure and the SCA method. Simulation results demonstrate that the proposed solution not only improves network throughput but also ensures stability in the UAV energy queue compared with baseline schemes. Future research directions include exploring more sophisticated mobility models and dynamic spectrum allocation, extending the framework to larger-scale networks to facilitate real-world implementations, and addressing perception and computation losses to enhance system performance.

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